HSLU Lucerne University of Applied Sciences and Arts

Chemical Reactions

CHEM: Chemistry

Davide Toniolo

Engineering and Architecture
October 20, 2026

FH Zentralschweiz

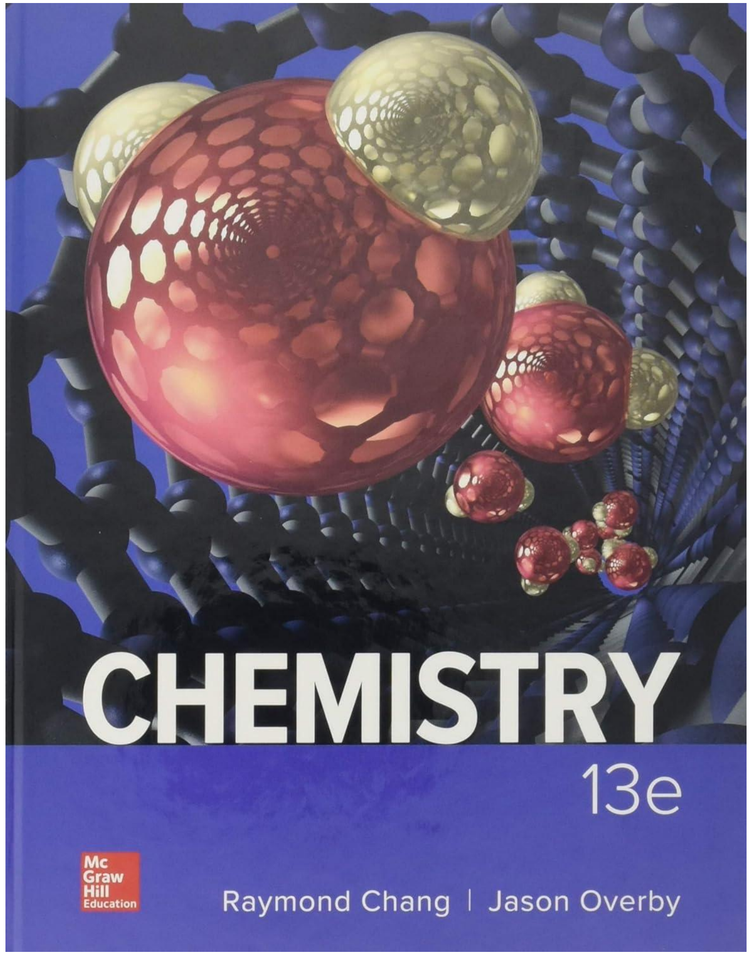
Course Details

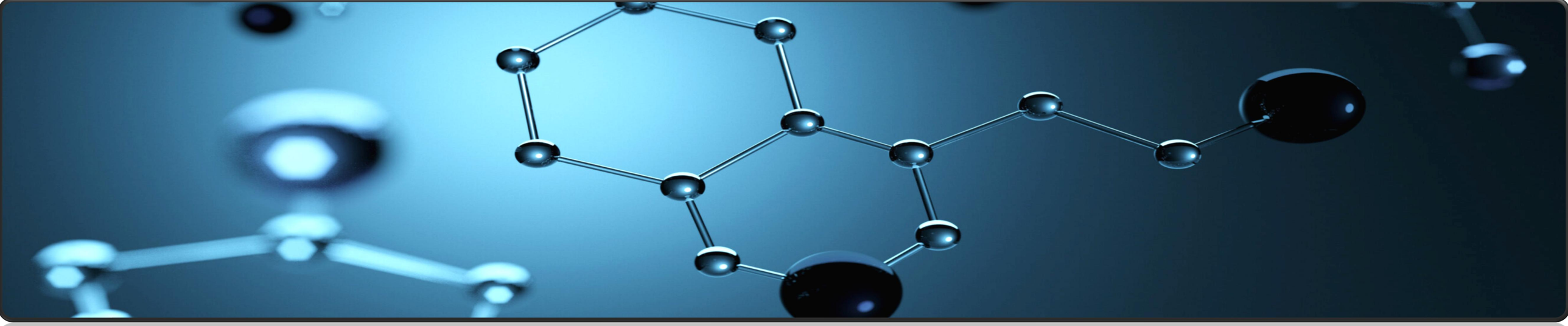
Lecturer	Date	Week	Topic
Macarena San Martin	16.09.2026	1	Introduction to Matter
Macarena San Martin	23.09.2026	2	Atomic Structure and Radioactivity
Macarena San Martin	30.09.2026	3	Atomic Models (Self Study), Periodic table and electronic configuration
Rebecca Ravotti	07.10.2026	4	Chemical Bonds and Hybridization
Rebecca Ravotti	14.10.2026	5	Lewis Structure
Davide Toniolo	21.10.2026	6	Chemical Reactions
Davide Toniolo	28.10.2026	7	Acids & Bases
Davide Toniolo	04.11.2026	8	Acids & Bases Exercises
Davide Toniolo	11.11.2026	9	Redox Reactions
Davide Toniolo	18.11.2026	10	Redox exercises
Rebecca Ravotti	25.11.2026	11	Metals and Semimetals
Rebecca Ravotti	02.12.2026	12	Organic Compounds nomenclature
Rebecca Ravotti	09.12.2026	13	Organic Exercises
Macarena /Rebecca/Davide	16.12.2026	14	Final Laboratory

Recommended literature

Chemistry ; Authors, Raymond Chang, Jason Overby ; Edition, **13, illustrated** ; Publisher, McGraw-Hill Education, 2018 ; ISBN, 1260085317, 9781260085310

Periodic table: **https://ptable.com**





Learning objectives

- Physical reaction
- Chemical reaction
- 1st and 2nd principles of thermodynamic
- Representing and balance chemical reactions
- Enthalpy, Entropy and Gibb’s free energy
- Chemical equilibrium
- Reaction kinetics

Prerequisites

- Chemical formulas
- Basic algebra

REACTIONS (IN CHEMISTRY)

Reaction are the processes trough which matter change.

The way a system reacts to a stimulus.

Physical reaction: brake or formation of secondary bonds (often including ionic and metallic)

- Change of state (solid-liquid-gas)
- Dissolution and mixing (salt in water)
- Change of size and shape (breaking a glass)

Chemical reaction: process involve the formation or breaking of primary chemical bonds (covalent).

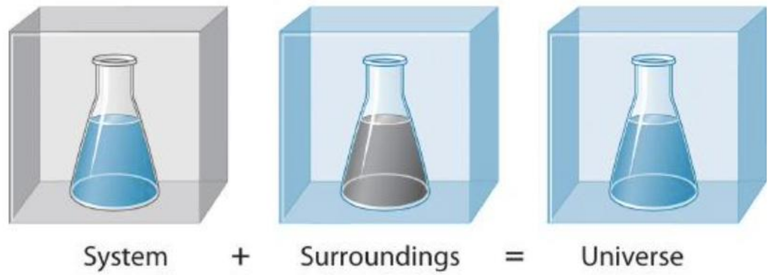
- Formation of new substances

REACTIONS: DEFINITION OF SYSTEMS

System: the portion of matter under study.

Ambient: everything which is not the system.

Universe: Ambient + System



- open system: exchange of matter and energy with surroundings



- closed system: only exchange of energy with surroundings



- isolated system: no exchange with surroundings (ideal)



PHYSICAL REACTIONS

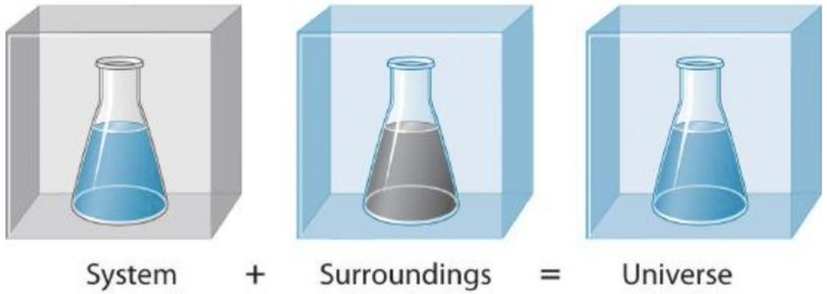
Physical reaction: process that alter the physical state of a system (shape size, state…) without changing the chemical composition (no change of covalent bonds)

In a physical system energy can assume two forms: heat and work

$$U_{univ} = \Delta U_{sys} + \Delta U_{surr} = 0$$

$$\Delta U_{sys} = -\Delta U_{surr}$$

$$\Delta U = Q - W$$

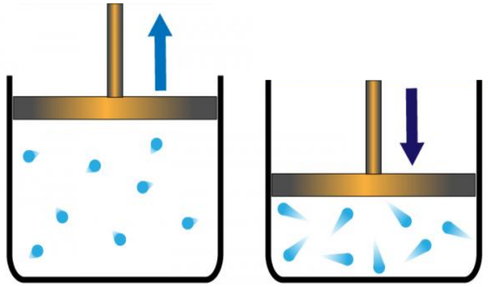


ΔU [J] = change in internal energy of the system. + if energy is gained, - if lost.
Q [J] = Energy transferred as heat. + if heat absorbed, - if released.
W [J] = Energy transferred as work. + if work done on the system, - if work is done by system to surroundings.

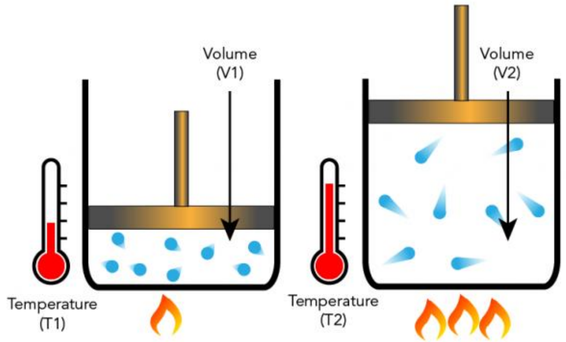
First law of thermodynamics: *“energy cannot be created or destroyed, only transferred or converted”*

GASSES: THE SIMPLEST SYSTEM

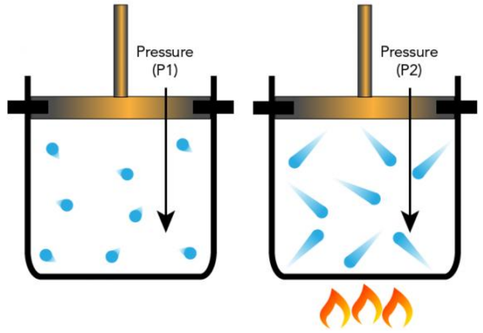
Gasses are the simplest (interesting) system to study because the reaction can be observed with a change of temperature, volume or pressure.



If V → ∞, P → 0
Heat is released to system,
T = constant
Boyle's Law: $P_1V_1 = P_2V_2$



If V → ∞, T → ∞
T must be increased to keep
P = constant
Charles's Law: $V_1/T_1 = V_2/T_2$



If T → ∞, P → ∞
For a system with
V = constant
Gay-Lussac's Law: $P_1/T_1 = P_2/T_2$

State equation of gasses:

$$PV = nRT$$

T [K]. → 1°C = 273.15 K
P [bar] → 1.013 bar = 1 atm
V [L]
n [mol]
R= 0.083 bar*K^{-1}*mol^{-1}

GASSES: EXERCISE

Scenario:
You trap 1.0 mol of N_2 gas in a piston at **25 °C** and **1.0 bar**.
The gas is compressed *isothermally* (T=constant) to **0.5 L** from an initial volume of **1.0 L**.

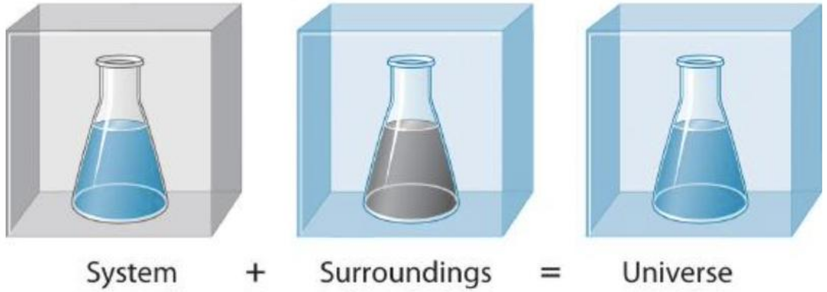
Tasks:
Check if everything is fine
1. Calculate the **final pressure**.
2. Determine whether **work is done on** or **by** the system.
2. Discuss: if temperature is constant, what happens to heat Q?
3. Bonus question: where did the work go?

Expected outcome:
1. P2=2.0 bar
2. Work is done **on** the gas (W > 0).
3. Because it's isothermal, **Q = –W** → heat flows *out* to surroundings.
4. The gas particles/atoms

CHEMICAL REACTIONS

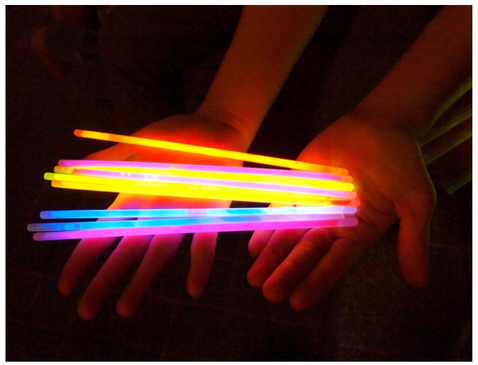
Chemical reaction: process in which one or more chemicals transform into different ones. It implies a change in the covalent bonds.

The system is defined in the same way.



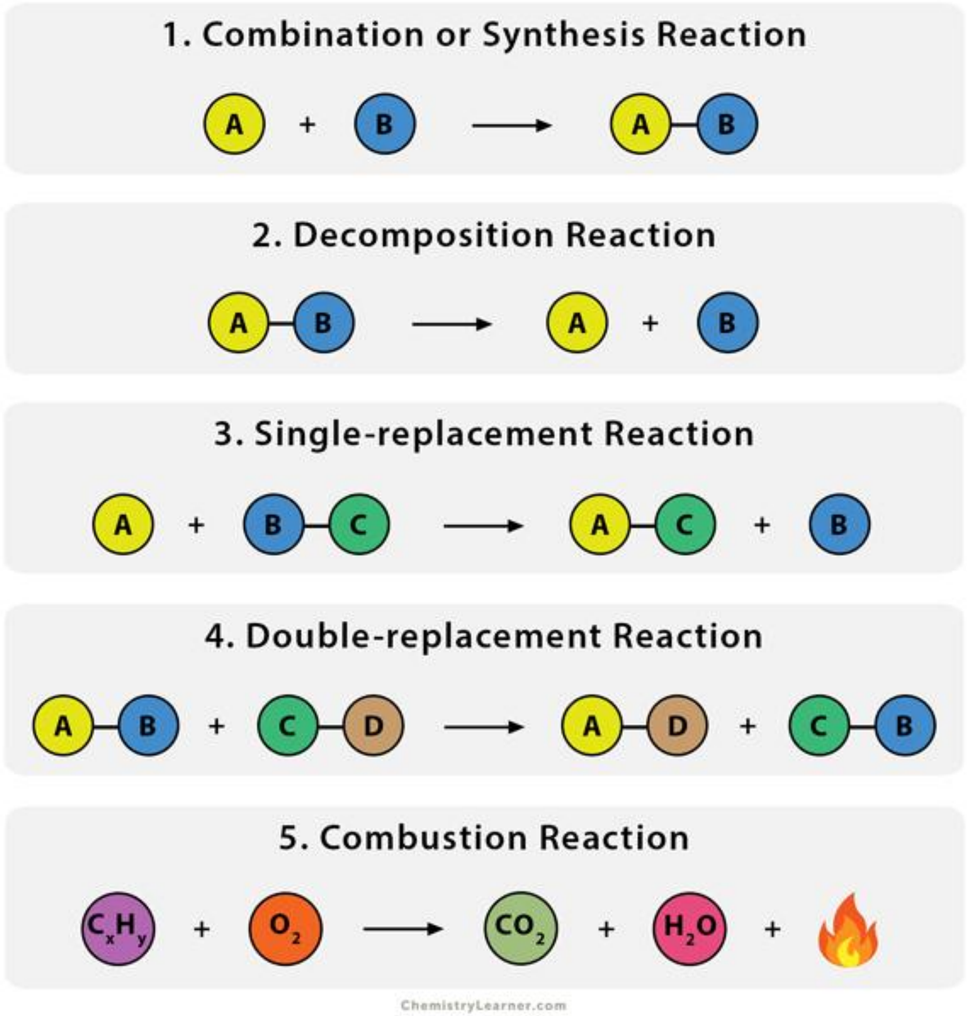
The first principle of thermodynamic is still valid, but new forms of energy are available.





CHEMICAL REACTIONS

Different kind of chemical reactions



Often associated with a heat transfer…

exothermic: release of heat

endothermic: absorption of heat

But other processed are possible:

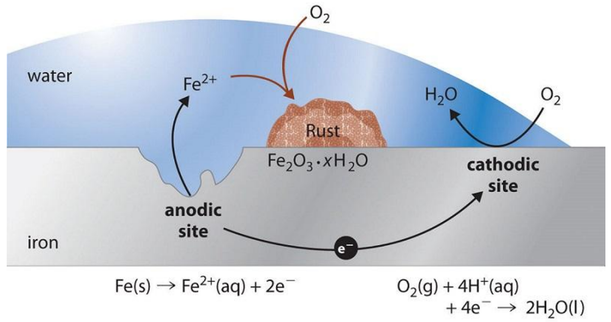
- Change of color
- Development of gas
- Formation of a precipitate
- Creation of a potential difference (current)
- Production of light
- …

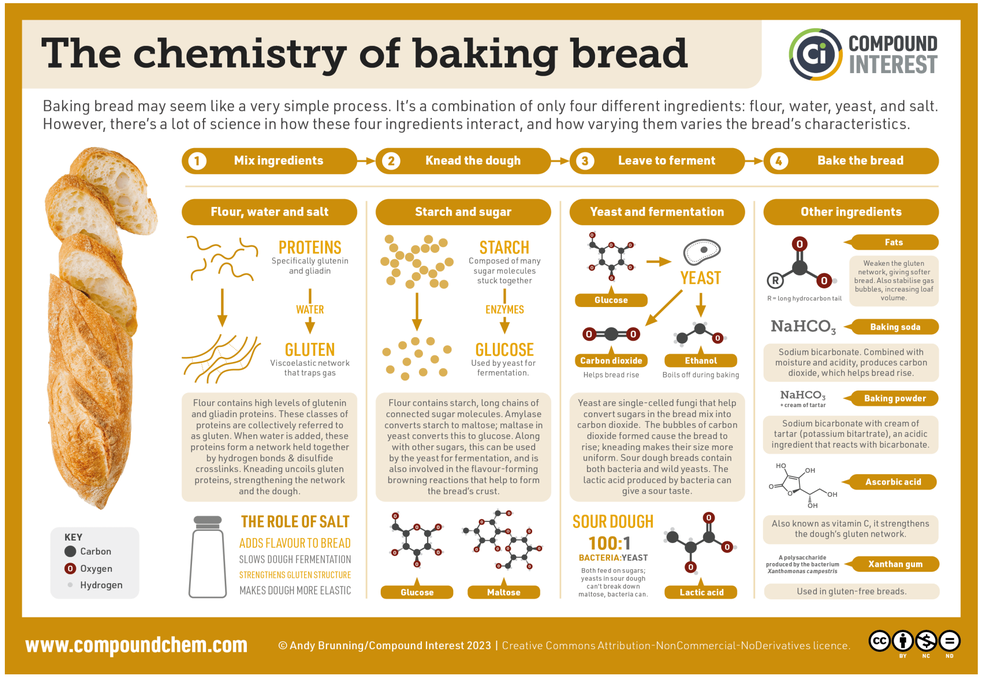
Which of the following are chemical reactions?

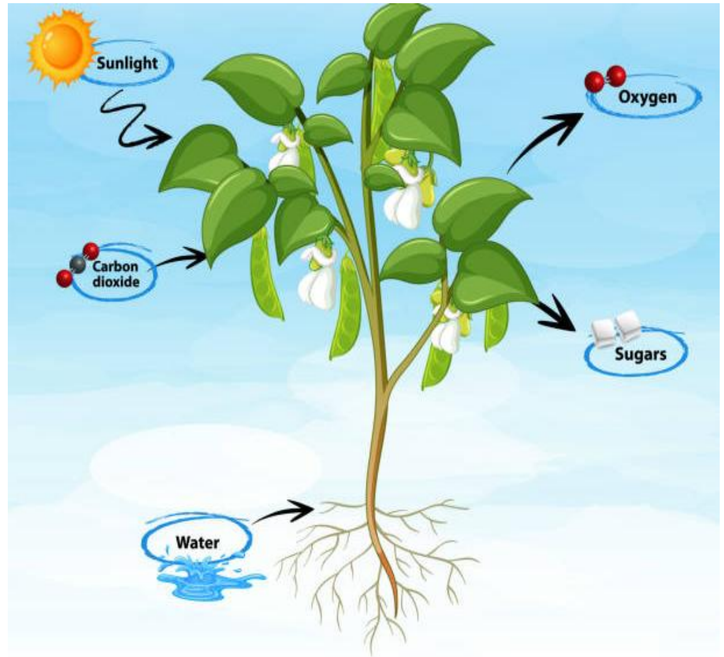
CHEMICAL REACTIONS

Everyday life chemical reactions





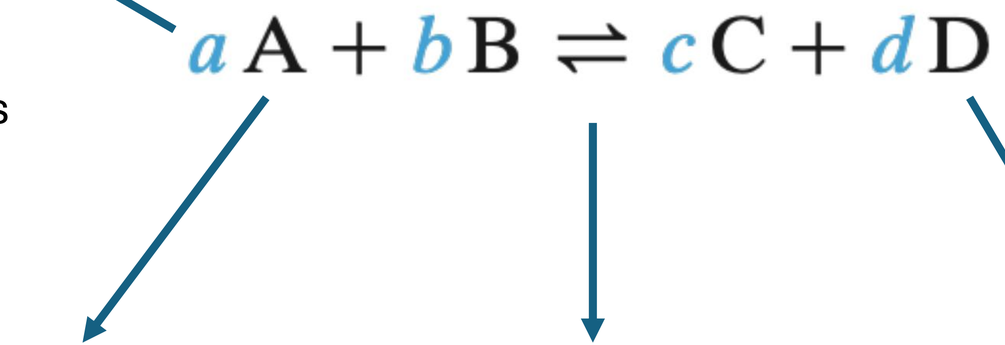




Other examples?

REPRESENTING CHEMICAL REACTIONS

Stoichiometric coefficient(s):
Indicate the amount of a specific reagent its ration with respect to the others



Reagent(s):
all the starting chemicals involved. Normally expressed with the chemical formula

Arrow(s):
Indicate the direction of the reaction in time

Product(s):
All the final chemical formed. Normally expressed with the chemical formula

BALANCING CHEMICAL REACTIONS

Balance a reaction means to determine the value of the stoichiometric coefficients

When the reagents and the products are known, the law of conservation of mass is sufficient to determine the stoichiometric coefficients (balance). The number of atoms of each element must be the same before and after the reaction.

1. Write the skeleton equation: Put the correct formulas for reactants on the left and products on the right, separated by an arrow.

2. Count the atoms (and changes): List the number of atoms for each element on both sides.

Treat an **unchanged polyatomic ion** (like SO_4^{2-}) as a single block.

3. Start with the most complex substance: Begin with the molecule that has the largest number of different atoms.

4. Balance elements one at a time:

Adjust coefficients for elements that appear in only one reactant and one product first.

Save free elements like pure metals or and elements like hydrogen and oxygen for later.

5. Check your work: Count every atom on both sides again to ensure they match completely. The charge must be balanced as well

BALANCING CHEMICAL REACTIONS: EXARCISE

Balance the following equations:

$N_2 + H_2 \rightarrow NH_3$

$Fe_2O_3 + CO \rightarrow Fe + CO_2$

$H_2SO_4 + NaOH \rightarrow Na_2SO_4 + H_2O$

$C_2H_6 + O_2 \rightarrow CO_2 + H_2O$

DOES A CHEMICAL REACTION HAPPEN?

Ammonia synthesis

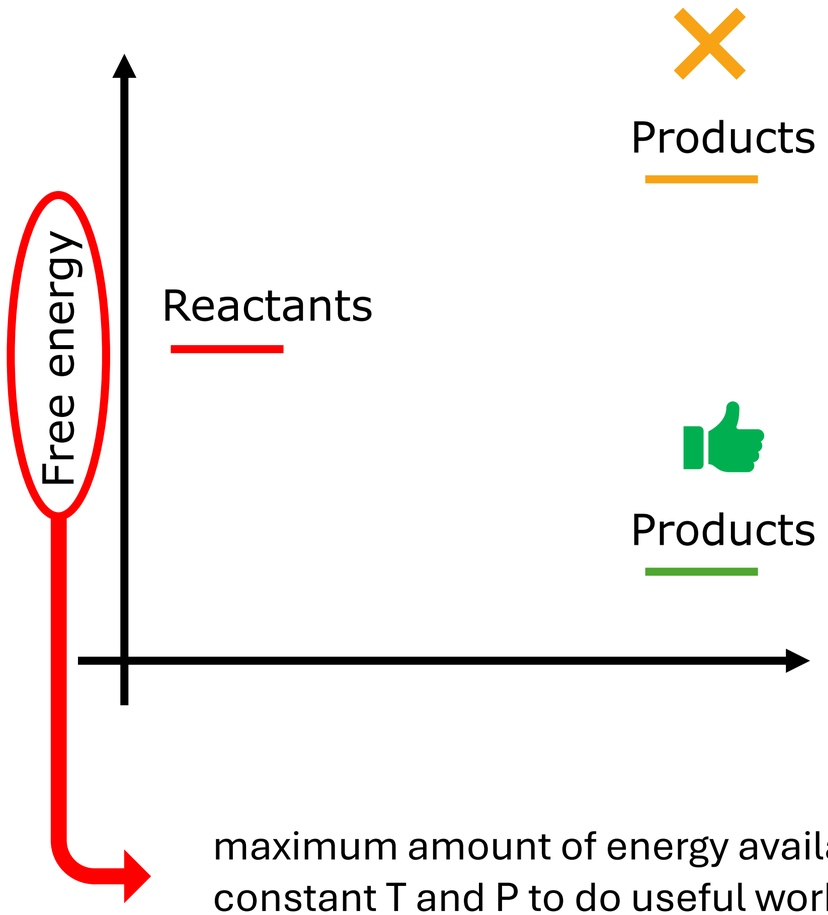
$N_2 + 3\ H_2 \rightarrow 2\ NH_3$

Why not: $2\ NH_3 \rightarrow N_2 + 3\ H_2$?

Every system tend towards the state that minimize its free energy

Second principle of thermodynamic:
"the entropy of an isolated system tent to increase"

Entropy is a concept connected with the degree of disorder of the system; it define the direction of a transformation



maximum amount of energy available in a system at constant T and P to do useful work
→ **Gibbs free energy**

GIBB'S FREE ENERGY

The values of Gibb's free energy variation indicate if a chemical reaction proceeds or not.

$$\Delta G = \Delta H - T\Delta S$$

ΔH = Enthalpy

$$\Delta H = \Delta U + \Delta PV$$

ΔH = Total heat content of the system at constant pressure

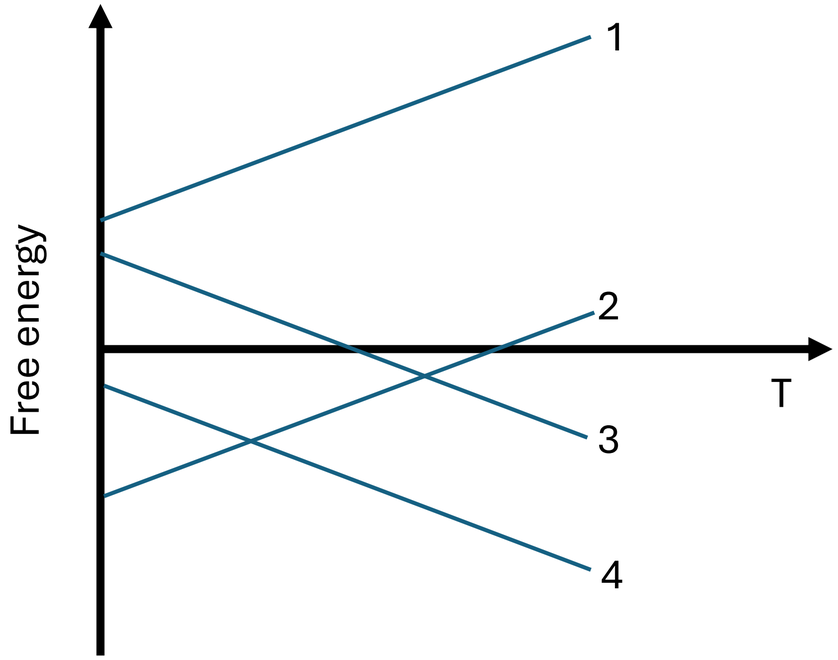
ΔS = Entropy

unavailability of thermal energy for conversion into mechanical work = degree of disorder or randomness in the system.

ΔS is connected with the degree of disorder of a system

GIBB'S FREE ENERGY

$$\Delta G = \Delta H - T\Delta S$$



	$\Delta H < 0$	$\Delta H > 0$
$\Delta S > 0$	Spontaneous at all T ($\Delta G < 0$)	Spontaneous at high T (when $T\Delta S$ is large)
$\Delta S < 0$	Spontaneous at low T (when $T\Delta S$ is small)	Non-spontaneous at all T ($\Delta G > 0$)

ENTHALPY AND ENTROPY

Molar enthalpy end entropy values are tabulated in the standard state (1 atm, 298 K).

Standand Enthalpies of Formation & Standard Entropies of Common Compounds

Substance	*State*	ΔH_f° ($\frac{kJ}{mol}$)	S° ($\frac{J}{mol \cdot K}$)
Ag	s	0	42.6
Ag^+	aq	105.79	72.7
AgCl	s	−127.01	96.2
AgBr	s	−100.4	107.1
$AgNO_3$	s	−124.4	140.9
Al	s	0	28.3
Al^{+3}	aq	−538.4	−321.7
$AlCl_3$	s	−704	110.7
Al_2O_3	s	−1675.7	50.9
Ba	s	0	62.8
$BaCl_2$	s	−858.6	123.7
$BaCO_3$	s	−1216.3	112.1
$Ba(NO_3)_2$	s	−992	214
BaO	s	−553.5	70.4
$Ba(OH)_2$	s	−998.2	112
$BaSO_4$	s	−1473.2	132.2
Br_2	ℓ	0	152.2
C	s	0	5.7
CCl_4	ℓ	−135.4	216.4
$CHCl_3$	ℓ	−134.5	201.7
CH_4	g	−74.8	186.2
C_2H_2	g	+226.7	200.8
C_2H_4	g	+52.3	219.5
C_2H_6	g	−84.7	229.5
C_3H_8	g	−103.8	269.9
CH_3OH	ℓ	−238.7	126.8
C_2H_5OH	ℓ	−277.7	160.7
CO	g	−110.53	197.6
CO_2	g	−393.51	213.6
CO_3^{-2}	aq	−675.23	−56.9
Ca	s	0	41.4
Ca^{+2}	aq	−543.0	−53.1
$CaCl_2$	s	−795.8	104.6
$CaCO_3$	s	−1206.9	92.9
CaO	s	−634.92	39.8
$Ca(OH)_2$	s	−986.1	83.4
$CaSO_4$	s	−1434.1	106.7
Cd	s	0	51.8
Cd^{+2}	aq	−75.92	−73.2
$CdCl_2$	s	−391.5	115.3
CdO	s	−258.35	54.8

Substance	*State*	ΔH_f° ($\frac{kJ}{mol}$)	S° ($\frac{J}{mol \cdot K}$)
Cl_2	g	0	223.0
Cl^-	aq	−167.080	56.5
ClO_4^-	aq	−128.10	182.0
Cr	s	0	23.8
Cr_2O_3	g	−1139.7	81.2
Cu	s	0	33.2
Cu^+	aq	+71.7	40.6
Cu^{+2}	aq	+64.8	−99.6
CuO	s	−157.3	42.6
Cu_2O	s	−168.6	93.1
CuS	s	−53.1	66.5
Cu_2S	s	−79.5	120.9
$CuSO_4$	s	−771.4	107.6
F^-	aq	−335.35	−13.8
F_2	g	0	202.7
Fe	s	0	27.3
$Fe(OH)_3$	s	−823.0	106.7
Fe_2O_3	s	−824.2	87.4
Fe_3O_4	s	−1118.4	146.4
H_2	g	0	130.6
H^+	aq	0	0.0*
HBr	g	−36.29	198.6
HCO_3^-	aq	−689.93	91.2
HCl	g	−92.31	186.8
HF	g	−273.30	173.7
HI	g	26.50	206.5
HNO_3	ℓ	−174.1	155.6
HPO_4^{-2}	aq	−1299.0	−33.5
HSO_4^-	aq	−886.9	131.8
H_2O	ℓ	−285.830	69.9
H_2O	g	−241.826	188.7
$H_2PO_4^-$	aq	−1302.6	90.4
H_2S	g	−20.6	205.7
Hg	ℓ	0	76.0
Hg^{+2}	aq	170.21	−32.2
HgO	cr,red	−90.79	70.3

*The standard entropy of the $H^+(aq)$ ion is defined to be 0.

$$\Delta H^\circ_{\text{reaction}} = \Sigma \Delta H^\circ_f(\text{products}) - \Sigma \Delta H^\circ_f(\text{reactants})$$

Example:

$Al + CO_2 \rightarrow Al_2O_3 + CO$

balance and calculate if the reaction is spontaneous

CHEMICAL EQUILIBRIUM

Every chemical reaction is in a dynamic equilibrium.

At any given interval of time, some reagents are transforming into products, and some products are transforming into reagents.

The value that describe the equilibrium is the equilibrium constant:

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

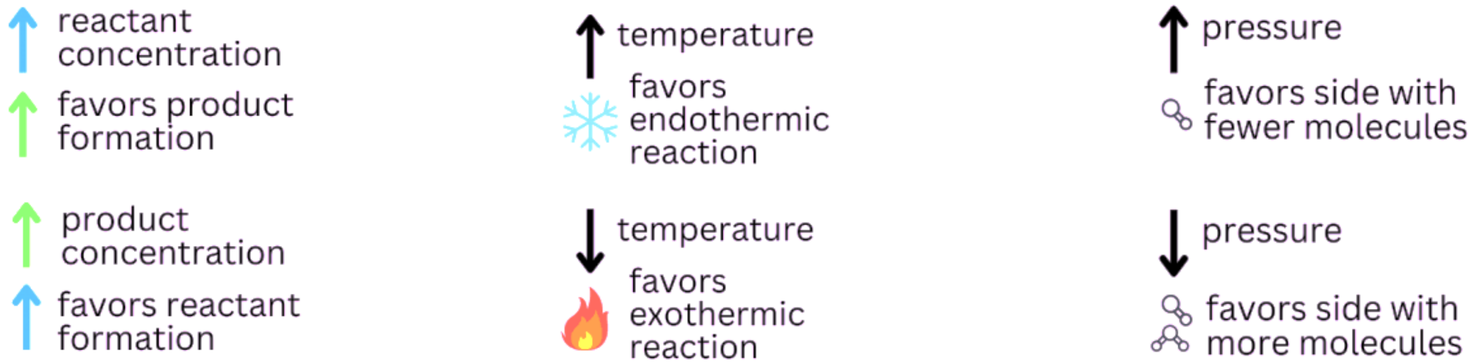
The equilibrium constant depends only on the temperature, and it is a condition that must be always verified.

CHEMICAL EQUILIBRIUM

Le Chatelier's principle: When a system is perturbated, it shift to counteract the change, reaching a new equilibrium state

$$a\,\mathrm{A} + b\,\mathrm{B} \rightleftharpoons c\,\mathrm{C} + d\,\mathrm{D}$$

The rules is valid also for "invisible" reaction products like heat or pressure



KINETIC

Gibb's free energy indicate if a reaction is spontaneous (happen) or not, but it gives no information about how long it takes.

Combustion of ethylene is spontaneous, but in absence of flame, the reaction do not happen.

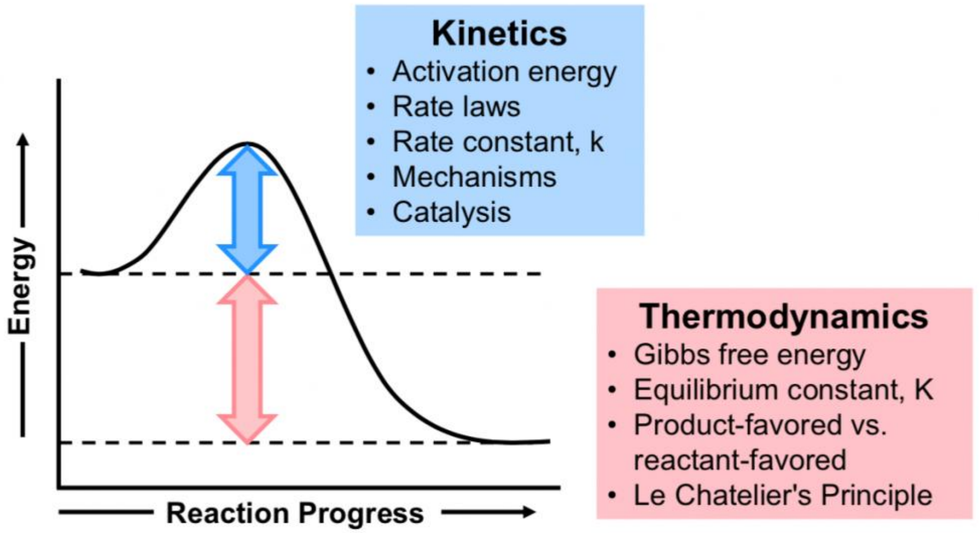
$$2\,C_2H_6 + 7\,O_2 \rightarrow 4\,CO_2 + 6\,H_2O$$

Every reaction have an activation energy to overcome to start.

Catalysts are substance able to decrease the activation energy of a chemical reaction

In absence of a catalyst a reaction can be speed up by:

- Increasing temperature
- Increasing the contact between reagents (pressure, concentration, surface area…)



SUMMARY

- Physical reaction

Every reaction that do not involve a change of substance

- Chemical reaction

Reaction in which chemicals are transformed

- 1st and 2nd principles of thermodynamic

1st: Energy cannot be created or destroyed, only transformed

2nd: The entropy of the universe tend to increase

- Representing and balance chemical reactions

Same number of atoms on both sides

- Enthalpy, Entropy and Gibb’s free energy

Indications if a reaction can spontaneously proceed.

- Chemical equilibrium

Concept of dynamic equilibrium, equilibrium constant

Discussion + Q&A





Outlook for Next Week:

- **Acids and bases**

Prerequisite

- Balancing chemical reactions
- Concept of equilibrium constant
- Basic math (quadratic equation, systems of equations)